The following formulas will be provided in the midterm test:

Some Standard Derivatives

1. $\frac{d}{dx}(x^n) = nx^{n-1}$
2. $\frac{d}{dx}(\cos x) = -\sin x$
3. $\frac{d}{dx}(\sin x) = \cos x$
4. $\frac{d}{dx}(\tan x) = \sec^2 x$
5. $\frac{d}{dx}(\sec x) = \sec x \tan x$
6. $\frac{d}{dx}(\csc x) = -\csc x \cot x$
7. $\frac{d}{dx}(\cot x) = -\csc^2 x$
8. $\frac{d}{dx}(e^x) = e^x$
9. $\frac{d}{dx}(\ln x) = \frac{1}{x}$
10. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
11. $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$
12. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
13. $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$
14. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$
15. $\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}}$

Some Standard Integrals

1. $\int (ax+b)^n \, dx = \frac{(ax+b)^{n+1}}{(n+1)a} + C \quad (n \neq -1)$
2. $\int \frac{1}{ax+b} \, dx = \frac{1}{a} \ln|ax+b| + C$
3. $\int e^{ax+b} \, dx = \frac{1}{a} e^{ax+b} + C$
4. $\int \sin(ax+b) \, dx = -\frac{1}{a} \cos(ax+b) + C$
5. $\int \cos(ax+b) \, dx = \frac{1}{a} \sin(ax+b) + C$
6. $\int \tan(ax+b) \, dx = \frac{1}{a} \ln|\sec(ax+b)| + C$
7. $\int \sec(ax+b) \, dx = \frac{1}{a} \ln|\sec(ax+b) + \tan(ax+b)| + C$
8. $\int \csc(ax+b) \, dx = -\frac{1}{a} \ln|\csc(ax+b) + \cot(ax+b)| + C$
9. $\int \cot(ax+b) \, dx = -\frac{1}{a} \ln|\csc(ax+b)| + C$
10. $\int \sec^2(ax+b) \, dx = \frac{1}{a} \tan(ax+b) + C$
11. $\int \csc^2(ax+b) \, dx = -\frac{1}{a} \cot(ax+b) + C$
12. $\int \sec(ax+b)\tan(ax+b) \, dx = \frac{1}{a} \sec(ax+b) + C$
13. $\int \csc(ax+b)\cot(ax+b) \, dx = -\frac{1}{a} \csc(ax+b) + C$
14. $\int \frac{1}{a^2+(x+b)^2} \, dx = \frac{1}{a} \tan^{-1}\left(\frac{x+b}{a}\right) + C$
15. $\int \frac{1}{\sqrt{a^2-(x+b)^2}} \, dx = \sin^{-1}\left(\frac{x+b}{a}\right) + C$
16. $\int \frac{-1}{\sqrt{a^2-(x+b)^2}} \, dx = \cos^{-1}\left(\frac{x+b}{a}\right) + C$
17. $\int \frac{1}{a^2-(x+b)^2} \, dx = \frac{1}{2a} \ln\left|\frac{x+b+a}{x+b-a}\right| + C$
18. $\int \frac{1}{(x+b)^2-a^2} \, dx = \frac{1}{2a} \ln\left|\frac{x+b-a}{x+b+a}\right| + C$
19. $\int \frac{1}{\sqrt{(x+b)^2+a^2}} \, dx = \ln\left|(x+b) + \sqrt{(x+b)^2+a^2}\right| + C$
20. $\int \frac{1}{\sqrt{(x+b)^2-a^2}} \, dx = \ln\left|(x+b) + \sqrt{(x+b)^2-a^2}\right| + C$
21. $\int \sqrt{a^2-x^2} \, dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1}\frac{x}{a} + C$
22. $\int \sqrt{x^2-a^2} \, dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2} \ln|x + \sqrt{x^2-a^2}| + C$

Some Trigonometric Identities

1. $\cos^2 x + \sin^2 x = 1, \qquad \sec^2 x - 1 = \tan^2 x, \qquad \csc^2 x - 1 = \cot^2 x$
2. $\sin A \cos A = \frac{1}{2} \sin 2A, \qquad \cos^2 A = \frac{1}{2}(1 + \cos 2A), \qquad \sin^2 A = \frac{1}{2}(1 - \cos 2A)$
3. $\sin A \cos B = \frac{1}{2}(\sin(A+B) + \sin(A-B)), \qquad \cos A \sin B = \frac{1}{2}(\sin(A+B) - \sin(A-B))$

   $\cos A \cos B = \frac{1}{2}(\cos(A+B) + \cos(A-B)), \qquad \sin A \sin B = -\frac{1}{2}(\cos(A+B) - \cos(A-B))$